

Dissertation / Project / Project Work Title:

Green Fuel Intelligence System (GFIS): A Physics-Guided AI Digital Twin for Methane Yield Prediction and Stability Monitoring in Anaerobic Digestion

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1. Broad Area of Work

The broad area of this dissertation is machine learning for sustainable energy systems, with a specific focus on physics-guided artificial intelligence for anaerobic digestion and biogas methane production. The work combines artificial intelligence, process engineering, time-series forecasting, explainable machine learning, soft sensing, digital twin simulation, optimization, and deployable software engineering.

The proposed system, Green Fuel Intelligence System (GFIS), aims to develop a research-grade AI framework that predicts methane yield from anaerobic digestion process variables and supports operational decision-making through stability monitoring, simulation, optimization, and physics-guided feasibility checking.

- Methane yield prediction for anaerobic digestion.
- Physics-guided machine learning for feasible prediction.
- VFA/ALK soft sensing and stability classification.
- Digital twin simulation for what-if process analysis.
- Explainable AI for identifying important process variables.
- API, dashboard, and browser-based demonstration for applied deployment.

2. Background

Anaerobic digestion is a biological process in which organic waste is converted into methane-rich biogas under oxygen-free conditions. Methane yield is the key performance indicator because it directly determines the energy recovery potential of the plant.

Methane production depends on interacting variables including temperature, pH, organic loading rate, hydraulic retention time, total solids, volatile solids, carbon-to-nitrogen ratio, moisture, ambient temperature, and prior reactor state. These variables interact in nonlinear and time-dependent ways, making prediction and optimization difficult.

Mechanistic models such as ADM1 provide a scientifically grounded representation of anaerobic digestion, but they are parameter-heavy and difficult to deploy directly in real-time decision-support software. Machine learning models such as XGBoost and LSTM can learn predictive relationships from operational data, but purely data-driven outputs may be physically unrealistic under unseen conditions.

GFIS addresses this gap by combining data-driven learning with physics-guided constraints. The system includes dataset generation, feature engineering, tabular prediction, time-series prediction, ensemble modeling, methane feasibility checking, VFA/ALK soft sensing, stability classification, what-if simulation, optimization, API services, dashboard interaction, and digital twin demonstration.

3. Objectives

- Design an anaerobic digestion dataset structure containing process variables, methane yield, VFA/ALK ratio, and stability labels.
- Build a data pipeline for missing-value handling, scaling, lag features, rolling statistics, interaction features, and LSTM-ready time windows.
- Develop an XGBoost-based tabular model for methane-yield prediction.
- Develop a PyTorch LSTM model for time-series methane-yield prediction.
- Construct an ensemble layer that can combine tabular and temporal model outputs.
- Incorporate a physics-guided methane-yield feasibility constraint based on volatile solids.
- Develop a VFA/ALK soft sensor and classify process stability as Stable, Warning, or Critical.
- Evaluate models using R2 score, RMSE, MAE, and physics violation count.
- Develop digital twin simulation capabilities for OLR sweep, stress scenarios, plant-run traces, and operating-point optimization.

- Expose the system through FastAPI endpoints, Streamlit dashboard, and browser-based digital twin interfaces.
- Prepare dissertation-ready reports, architecture explanation, demo evidence, and presentation material.

4. Scope of Work

The scope is to independently design and implement GFIS as a working software and machine-learning prototype for physics-guided methane-yield prediction in anaerobic digestion. The work includes dataset design, preprocessing, model training, physics-guided correction, soft sensing, simulation, optimization, evaluation, reporting, and deployment planning.

- Data generation and preprocessing using Python, Pandas, NumPy, and Scikit-learn.
- Synthetic and literature-inspired anaerobic digestion dataset creation for controlled experiments.
- Feature engineering using lag features, rolling means, process ratios, interaction terms, and methane-history variables.
- XGBoost model development for nonlinear tabular prediction and PyTorch LSTM development for temporal modeling.
- Ensemble prediction and physics-guided correction using a volatile-solids methane feasibility bound.
- VFA/ALK soft sensor, stability classification, feature importance, and SHAP-ready explainability.
- Controlled research scenarios including nominal operation, overload, pH shock, low-temperature disturbance, reduced HRT, high-solids feedstock, and corrective recovery.
- FastAPI backend, SQLite persistence, Streamlit dashboard, HTML digital twin, Docker/Ubuntu deployment notes, and future ROS2/Gazebo integration planning.

The current semester work will not involve direct deployment on an industrial biogas plant. The prototype will be validated using synthetic and literature-inspired data, with the architecture kept ready for future calibration using public, laboratory, IoT, SCADA, or plant-specific datasets.

5. Plan of Work

Phases	Start Date - End Date	Work to be done
Dissertation Outline	5 May 2026 - 10 May 2026	Finalize problem statement, abstract, project layout, scope, supervisor alignment, and BITS outline report.
Literature Review and Data Pipeline	11 May 2026 - 7 June 2026	Review ADM1, methane-yield prediction, VFA/ALK stability, physics-guided ML, digital twins, and soft sensors. Build dataset generator and preprocessing pipeline.
Model Development	8 June 2026 - 5 July 2026	Develop baseline models, XGBoost tabular model, PyTorch LSTM model, feature engineering, model saving, and evaluation pipeline.
Physics-Guided Ensemble and Soft Sensor	6 July 2026 - 2 August 2026	Build ensemble prediction, methane feasibility constraint, VFA/ALK soft sensor, stability classification, and explainability artifacts.
Digital Twin, API, and Dashboard	3 August 2026 - 30 August 2026	Implement FastAPI services, Streamlit dashboard, HTML digital twin, what-if simulation, plant-run trace, optimization, and stress-test experiments.
Evaluation and Final Submission	31 August 2026 - 13 September 2026	Prepare evaluation report, research scenario report, screenshots, final dissertation report, presentation, demo evidence, and final submission artifacts.

6. Literature References

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7. Particulars of the Supervisor and Additional Examiner

Particular	Supervisor	Additional Examiner
Name	Dr. Dipti Jadhav	Navnath Krishnath Mane
Qualification	Ph.D. (Computer Engineering), M.E. (Information Technology), B.E. (Information Technology)	M.E. (Computer Science and Engineering), B.E. (Information Technology), Ph.D. (Pursuing)
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8. Remarks of the Supervisor

To be completed and signed by the supervisor after review.

Signature of Student: _____

Name: Akash Shivadas Chatake

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Date: _____

Signature of Additional Examiner: _____

Name: Navnath Krishnath Mane

Date: _____